

Adarsh Anand

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About

Software engineer specializing in distributed systems, AI-assisted engineering, and cloud infrastructure. Experienced in large-scale data processing, microservices, infrastructure automation, and compliance-sensitive domains, including identity governance at Oracle and US healthcare technology at Blink Health. Uses AI as an engineering multiplier while maintaining ownership of system behavior, correctness, and reliability.

Education

Indian Institute of Technology Kanpur

B.Tech in Electrical Engineering

Kanpur, Uttar Pradesh

2019 – 2023

Work Experience

Blink Health

Senior Software Engineer – Prescription Processing

Bengaluru, India

Jun 2026 – Aug 2026

- Led **end-to-end migration of Automated Drug Reviews from a monolithic architecture to microservices**, decomposing **7 critical service-to-service (S2S) calls** into 7 independent execution paths across **3 new services**, preserving correctness across critical prescription workflows. (*Python, AWS, Amazon SQS, PostgreSQL, NoSQL, Terragrunt*)
- Built a **repository visualization and code-flow analysis tool** that mapped a **500K-line monolith into 50+ codeflows**, tracing prescription lifecycle transitions, dependencies across **40+ database tables**, object relationships, and **40+ API endpoints** documented in a Postman collection; recognized during the organization-wide **AI adoption meeting**.
- Created **3 organization-wide AI engineering skills** for closed-loop development and testing of simple bugs, combining code changes, test execution, and validation into repeatable workflows.
- Constructed a **New Relic telemetry dashboard using NRQL** to monitor **25 S2S calls** across critical prescription-processing workflows, improving visibility into service behavior, operational health, and potential failure points.
- Applied AI-assisted engineering workflows using **Claude Code, Cursor, OpenAI Codex, Rovo AI, and MCP integrations** for codebase exploration, implementation, debugging, and automated testing.

Oracle

Member of Technical Staff IC2 – Access Governance

Hyderabad, India

Jun 2023 – May 2026

- Designed **40+ SQL queries** for an interconnected Access Governance dashboard covering entitlements, roles, and user activity across **18 tables and 100+ indexes**; reduced query latency by **6.5×** through AWR-guided tuning and partitioning.
- Automated deployment of Docker containers for **50+ microservices across 15+ Kubernetes namespaces** using Helm charts and shell scripts, reducing deployment time by **60%+**. (*Docker, Kubernetes, Helm, Zsh*)
- Optimized the logging pipeline by removing **39% of redundant data (12 GB to 7.3 GB/day)** using regular expressions and improving processing speed by **4×** through log-processing and truncation optimizations. (*OCI Logging, Regex, Groovy*)
- Transformed **10 TB+ of legacy data from 30+ targets** using JOLT specifications and Parquet formatting; implemented Redis caching to reduce API latency by **65%** while achieving a **95% cache-hit rate**. (*ETL, Redis, Parquet*)
- Engineered the Direct Modify Account feature by combining TMS API responses, static configuration, and database data into **JSON Schema-compliant payloads**, enabling UI-level editing of complex nested attributes and resolving **three-year-old technical debt**. (*REST APIs, JSON Schema*)
- Spearheaded the end-to-end development of Oracle Analytics Cloud (OAC) **proof-of-concept reports** for CorpIDM onboarding under Audit & Compliance requirements. (*OAC, SQL*)
- Defined report architecture and development guidelines adopted by the team, delivered knowledge-transfer sessions to **4 senior engineers**, and collaborated with the **Chief Architect** to finalize requirements and report design. (*Audit & Compliance*)

Oracle

Intern – Oracle Content Management (OCM)

Bengaluru, India

May 2022 – Jul 2022

- Conceptualized a software to prevent **cross-site scripting (XSS)** attacks using 200+ Regex patterns over **100+ GB** of files uploaded to Oracle Content Management.
- Received a **Pre-Placement Offer (PPO)** following the internship in the OCM division.

Leadership Experience

Placement Office, IIT Kanpur

Kanpur, Uttar Pradesh

Placement Coordinator

Mar 2021 – Feb 2022

- Directed a team of **70+ company coordinators and 100+ volunteers** during IIT Kanpur's internship drive for **900+ students**; helped architect a unified career portal for **1,400+ candidates**, reducing administrative overhead by **40%**.
- Forged partnerships with **50+ Fortune 500/GCC firms**, including Google, Microsoft, and Optiver, and maintained a corporate database tracking **1,000+ companies** through stakeholder coordination and campus recruitment events.

Key Projects

Local AI Environment – M

Personal Project | Open WebUI, LM Studio, Gemma 4

2026

- Deployed a **local agentic workflow** using **Docker** containers, integrating **Open WebUI** with **LM Studio** for private, offline LLM orchestration.
- Hosted **Gemma 4** locally using **Q4_K_M quantization** to reduce model memory requirements while retaining practical local inference capabilities.

Gold-Denominated World Assets GDWA – Q

Personal Project | Python, Streamlit, Plotly, Pandas, yFinance

2025

- Built a financial analytics dashboard comparing **10+ global assets** (S&P 500, Nifty 50, Nikkei, BTC, ETH, and Silver) against gold (XAU) to analyze purchasing-power performance beyond fiat denomination.
- Implemented a multi-currency denomination engine, incremental data synchronization, rolling 30-day volatility, drawdown heatmaps, and asset-correlation matrices.

Algorithmic Trading & Backtesting Suite – Q

Personal Project | Python, Go, Jupyter, Binance API

2024

- Developed a **full-stack trading toolkit** comprising a **strategy backtester**, **live crypto mock-trading engine**, **real-time market data fetcher** via Binance WebSocket, and an **AI-model-driven strategy** module.
- Built a live data pipeline in **Go** for order-book tracking, with a Python layer for strategy execution, signal generation, and performance analytics across equities and crypto.

VALM – Vedic Astrology Life Map – Q

Personal Project | Python, Flask, Swiss Ephemeris, JavaScript, SVG

2026

- Built a **full-stack Vedic astrology application** using Swiss Ephemeris for sidereal planetary calculations with Lahiri ayanamsa, generating birth charts from date, time, timezone, and geographic coordinates.
- Engineered an **astrological interpretation engine** encoding classical rules for planetary strength, exaltation, debilitation, retrograde states, conjunctions, yogas, Nakshatras, and career and wealth indicators.
- Developed interactive **North and South Indian birth charts**, a **360-degree ecliptic wheel**, and a **planetary transit timeline** using SVG and vanilla JavaScript, enabling users to explore planetary positions and transits over time.

Technical Skills

Languages: C, C++, Java, Python, Go, JavaScript, TypeScript, PL/SQL

Backend & Distributed Systems: REST APIs, Microservices, Redis, Kafka, Flink, Amazon SQS

Databases & Data Processing: PostgreSQL, MongoDB, NoSQL, SQL, Parquet, ETL

Cloud & Infrastructure: AWS, Kubernetes, Docker, Helm, Terragrunt, CI/CD, GitHub Workflows

Observability: New Relic, NRQL, Grafana, OCI Logging

AI Engineering: Claude Code, Cursor, OpenAI Codex, RoVo AI, MCP, local LLMs, quantization

Extracurricular Activities

- * **Toastmasters International, IIT Kanpur:** Helped develop public-speaking skills for **100+ students** through **20+ group discussions and speaking sessions** as a founding member of the first IIT Kanpur Toastmasters chapter.